

Deep Learning Foundations

Chapter 7: Gradient Descent

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7.14)

Solution:

To show that the renormalized variables have zero mean:

$$\begin{aligned}\tilde{\mu}_i &= \frac{1}{N} \sum_{n=1}^N \tilde{x}_{ni} && \text{(definition of the mean of } \tilde{x}) \\ &= \frac{1}{N} \sum_{n=1}^N \frac{x_{ni} - \mu_i}{\sigma_i} && \text{(substitute definition of } \tilde{x}_{ni}) \\ &= \frac{1}{N\sigma_i} \left(\sum_{n=1}^N x_{ni} - \sum_{n=1}^N \mu_i \right) && \text{(factor out } \sigma_i \text{ and distribute)} \\ &= \frac{1}{N\sigma_i} \left(\sum_{n=1}^N x_{ni} - N\mu_i \right) && \text{(since } \sum_{n=1}^N \mu_i = N\mu_i) \\ &= \frac{1}{N\sigma_i} (N\mu_i - N\mu_i) && \text{(use } \sum_{n=1}^N x_{ni} = N\mu_i) \\ &= 0\end{aligned}$$

To show that the renormalized variables have unit variance:

$$\begin{aligned}\tilde{\sigma}_i^2 &= \frac{1}{N} \sum_{n=1}^N (\tilde{x}_{ni} - \tilde{\mu}_i)^2 && \text{(definition of variance for } \tilde{x}) \\ &= \frac{1}{N} \sum_{n=1}^N \tilde{x}_{ni}^2 && \text{(since } \tilde{\mu}_i = 0) \\ &= \frac{1}{N} \sum_{n=1}^N \left(\frac{x_{ni} - \mu_i}{\sigma_i} \right)^2 && \text{(substitute definition of } \tilde{x}_{ni}) \\ &= \frac{1}{N\sigma_i^2} \sum_{n=1}^N (x_{ni} - \mu_i)^2 && \text{(distribute power and factor out } \sigma_i^2) \\ &= \frac{1}{\sigma_i^2} \left(\frac{1}{N} \sum_{n=1}^N (x_{ni} - \mu_i)^2 \right) && \text{(rearrange terms)}\end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\sigma_i^2} \cdot \sigma_i^2 && \text{(definition of } \sigma_i^2 \text{ from 7.49)} \\
 &= 1 && \text{(simplify)}
 \end{aligned}$$